Starter Task: Sumcheck-Based ZKML Proving for ResNet-18 on CIFAR-10

Goal

The goal of this starter task is to assess your ability to implement and reason about a concrete sumcheck-based ZKML pipeline for neural-network inference.

You will prove **ResNet-18 inference on CIFAR-10** in a **zkCNN-like architecture**, using the following repository as the reference implementation:

https://github.com/TAMUCrypto/zkCNN

The main technical focus should be on supporting **BatchNorm** and **residual connections** correctly.

Task

Build a runnable implementation that proves ResNet-18 inference on at least one CIFAR-10 image in a zkCNN-like/sumcheck-based proving architecture.

The existing zkCNN repo may be used as a starting point, but ResNet-18 support will require additional implementation work, especially for:

1. **BatchNorm**
2. **Residual connections**
3. The ResNet-18 computation graph for CIFAR-10

The implementation does not need to be production-quality, but it should be complete enough to demonstrate an end-to-end proof flow for at least one CIFAR-10 input.

Technical Focus

BatchNorm

BatchNorm is the main focus of this task. The solution should explain how BatchNorm is represented in the arithmetization and why the resulting construction is sound.

Residual Connections

ResNet-18 also contains residual connections. The solution should explain how these are handled in the sumcheck-based proving flow.

In particular, residual connections may create multiple opening claims for the same input witness. The writeup should explain how these claims are managed so that downstream claims do not grow exponentially.

Expected Outputs

Please submit:

1. **Runnable implementation**
    - Code that proves ResNet-18 inference on at least one CIFAR-10 image.
    - Clear build/run instructions.
    - A correctness check against plain PyTorch or NumPy inference.
2. **Short technical writeup**
    - How ResNet-18 is represented in the zkCNN-style architecture.
    - How BatchNorm is handled and why it is sound.
    - How residual connections are handled.
    - Any assumptions, limitations, or unsupported cases.
3. **Basic profiling numbers**
    - Prover time.
    - Verifier time.
    - Proof size.
    - Optional: layer-level cost breakdown if easy to collect.